

ST-MGPrompt

Overall Framework

History



Wind-Farm History

134 turbines
16 variables
144 steps = 24 h
 $B \times 144 \times 134 \times 16$

16 → 64

shared projection

1 Dual Temporal Semantics

same 144-step grid

Fine



local variation

causal conv., $k=3$

Coarse



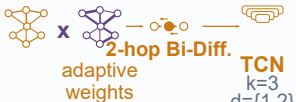
trend context

windows 6 / 18 / 36

2 Graph-Temporal Encoding

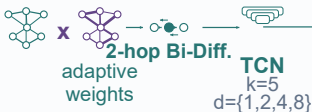
Micro

dWspd + distance



Macro

Patv trend + distance



3 Cross-Granularity Fusion

Macro-Trend Prompt

P1 P2 P3 P4

4 tokens / turbine

trend guidance



Fine history

H'f



Coarse history

H'c

recent-detail correction

4 Task-Conditioned Forecasting

Last-State Context

Fine last
Coarse last
Macro mean



node / step
fixed type

Node-Horizon ST Prompt

Direct Decoder

10-step forecast

$B \times 10 \times 134$

H3 H6 H10

Training Only - MS-MG-DWU

$\hat{y}$ at, Y, M

Horizon difficulty
+ Site difficulty

Objective

not used in inference